

CAIRO INTL, Egypt

WMO: 623660

Lat: 30.122N

Lon: 31.406E

Elev: 382

StdP: 14.49

Time Zone: 2.00 (E02)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	46.7	48.6	24.6	19.0	69.0	28.6	22.9	68.7	27.0	60.9	23.6	60.9	5.7	90	0.423

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	18.6	101.8	69.9	98.8	70.6	96.7	70.9	77.6	88.9	76.5	87.9	75.6	86.9	11.6	350

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
75.0	133.1	82.4	73.5	126.6	81.6	72.2	120.9	81.0	41.4	89.2	40.3	87.8	39.4	87.2	82.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 21.1	18.4	16.3	DB	43.1	108.2	3.2	3.3	40.8	110.5	38.9	112.4	37.1	114.3	34.7	116.6	(4)
(5)			WB	38.5	80.0	2.8	1.5	36.5	81.0	34.8	81.9	33.2	82.7	31.2	83.8	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	73.3	58.5	60.8	65.6	71.9	78.7	83.2	85.1	85.4	82.7	76.8	68.6	61.2	(6)
(7)		DBStd	10.64	3.83	5.02	6.09	6.40	5.38	4.01	2.92	2.53	3.20	4.20	4.49	3.94	(7)
(8)		HDD50	1	1	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	560	206	138	63	7	0	0	0	0	0	0	16	130	(9)
(10)		CDD50	8491	264	303	483	656	888	995	1089	1098	980	830	557	348	(10)
(11)		CDD65	3574	4	20	81	213	423	545	624	633	530	365	123	13	(11)
(12)		CDH74	40148	30	176	751	2176	4734	6709	7957	8042	5914	2996	608	55	(12)
(13)		CDH80	20057	2	48	282	995	2426	3684	4313	4224	2862	1091	122	8	(13)
(14)	Wind	WSAvg	8.2	7.7	8.3	9.1	9.7	9.4	9.0	8.1	7.7	8.1	7.6	6.9	7.3	(14)
(15)	Precipitation	PrecAvg	0.9	0.2	0.2	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.1	0.2	(15)
(16)		PrecMax	1.8	0.9	0.6	1.2	0.4	0.2	0.0	0.0	0.0	0.0	0.6	0.8	1.0	(16)
(17)		PrecMin	0.2	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	0.0	(17)
(18)		PrecStd	0.4	0.2	0.2	0.2	0.1	0.0	0.0	0.0	0.0	0.0	0.1	0.2	0.3	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	77.8	86.3	95.0	102.4	105.6	105.6	104.0	100.7	102.2	96.8	89.0	80.0	(19)
(20)			MCWB	56.3	60.8	61.5	66.2	66.5	69.6	72.7	73.1	70.3	68.2	64.4	60.3	(20)
(21)		2%	DB	72.1	79.2	87.4	95.1	98.9	100.6	98.9	98.4	97.1	91.7	82.8	74.1	(21)
(22)			MCWB	55.1	57.6	60.6	64.2	66.7	70.3	72.8	73.3	71.3	68.8	64.3	59.2	(22)
(23)		5%	DB	68.6	74.0	82.1	89.9	95.2	97.2	96.8	96.5	93.8	88.7	80.1	71.5	(23)
(24)			MCWB	54.9	56.3	59.0	62.7	66.1	70.0	72.8	73.4	71.6	68.9	63.8	58.4	(24)
(25)		10%	DB	66.4	70.0	77.1	85.7	91.4	94.7	94.8	94.6	91.5	85.9	77.1	69.1	(25)
(26)			MCWB	54.3	55.1	58.4	61.7	65.7	70.1	72.7	73.3	71.4	68.5	62.9	57.2	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.7	63.3	67.1	69.5	71.6	76.4	79.0	79.6	77.4	74.9	70.6	64.7	(27)
(28)			MCD B	69.9	78.8	82.8	90.2	92.1	91.2	91.0	90.1	86.7	83.8	78.7	72.0	(28)
(29)		2%	WB	58.5	60.3	63.3	66.7	69.8	74.6	77.3	78.1	75.7	72.7	68.5	62.2	(29)
(30)			MCD B	66.7	72.7	79.7	86.8	87.9	88.7	89.4	89.0	86.3	82.5	76.4	70.8	(30)
(31)		5%	WB	56.9	58.3	61.3	64.8	68.6	73.2	76.1	77.0	74.6	71.5	66.6	60.4	(31)
(32)			MCD B	65.7	69.3	75.6	84.5	86.3	88.6	88.4	88.0	85.5	81.0	75.1	68.6	(32)
(33)		10%	WB	55.6	56.7	59.6	63.0	67.4	72.1	75.2	76.0	73.5	70.3	64.9	58.7	(33)
(34)			MCD B	64.0	67.2	72.9	80.5	84.8	87.1	87.5	86.9	84.6	80.3	73.9	66.6	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	15.5	16.6	18.6	21.3	22.5	21.5	19.6	18.6	18.5	17.2	16.4	15.4	(35)
(36)			MCDBR	18.5	22.5	25.1	27.9	26.9	24.8	21.8	20.5	21.2	20.2	19.1	17.8	(36)
(37)			MCWBR	9.4	9.6	9.6	9.9	9.0	7.4	7.0	6.6	6.5	6.5	7.9	9.0	(37)
(38)		5% WB	MCDBR	16.9	19.3	22.0	25.1	23.6	22.3	20.4	19.2	19.6	18.3	17.0	16.6	(38)
(39)			MCWBR	9.5	9.0	9.5	9.7	7.9	7.0	6.8	6.4	6.2	6.2	7.7	9.3	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.442	0.490	0.537	0.569	0.575	0.487	0.474	0.476	0.483	0.494	0.432	0.418	(40)	
(41)		taud	1.966	1.845	1.736	1.693	1.710	1.946	1.994	1.990	1.962	1.934	2.069	2.067	(41)	
(42)		Ebn at Noon	240	240	239	237	237	257	260	258	251	236	243	242	(42)	
(43)		Edh at Noon	49	60	71	77	76	60	57	56	56	54	43	42	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1137	1402	1828	2167	2411	2607	2544	2361	2047	1614	1246	1060	(44)	
(45)		RadStd	35	64	58	69	77	28	19	26	34	39	42	29	(45)	

Historical Trends

		Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)
(46) Station Only	DBAvg	+1.08	N/A	-2.45	N/A	N/A	N/A	-90	+382	+308
(47) Regional (6 neighbors)	DBAvg	+0.99	N/A	-1.17	+0.95	+0.67	N/A	-104	+358	+254

Nomenclature: See separate page